

PEAK ACHIEVERS

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GRADE 11 SENIOR SCHOOL PHYSICS

BOOK 2 – REVISION & ASSESSMENT

● Topical Quizzes ● Mixed Exercises ● Practice Tests ● Answers & Marking Guidance



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PEAK ACHIEVERS | GRADE 11 PHYSICS

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Peak Achievers Online Academy
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Peak Achievers Grade 11 Senior School Physics — Book 2: Revision & Assessment is an independently developed supplementary learning resource intended to support Grade 11 Physics within Kenya's Competency-Based Education framework. It is not an official publication of KNEC or KICD, and the practice assessments in this book should not be interpreted as official examination papers.

First Edition: 2026

Dedication

This book is dedicated to the loving memory of my late father, Simeon Nyatuka, whose life, guidance and sacrifices remain a lasting source of inspiration. To my mother, Hyrine Kerubo; my guardians, Peter Munde Mong'are and Teresa Kwamboka Sawe; my friends Mnaree, Richie, Duke, Caleb, Cuba, Philo, Rob, Tim, Mr Peter Kimani and many others; to Her; to my fellow teachers; and to every learner determined to improve.

— Hezekiah Omweri Nyatuka

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Preface

Physics is mastered by understanding principles, applying them to unfamiliar situations, interpreting evidence, carrying out investigations and learning from mistakes. Book 2 is the revision and assessment companion to Book 1: Concise Notes. It follows the Grade 11 Senior School Physics scope used throughout the Peak Achievers series and emphasises application, inquiry, data handling, graph work and scientific reasoning.

How to Use This Book

REVISE → PRACTISE → APPLY → ASSESS → CHECK → MASTER

- Read the Quick Revision section before attempting each topical quiz.
- Attempt all questions before checking the marking guidance.
- Use graph grids and drawing spaces carefully: label quantities, units and scales.
- After the 13 topical chapters, attempt the five Mixed Revision Exercises.
- Sit the five Final National-Assessment-Style Practice Tests under timed conditions.
- Use the answers to diagnose errors and retry difficult questions.

Command Words

Command	Expected response
State/Identify	Give a concise fact or name.
Explain	Give the Physics reason or mechanism.
Calculate/Determine	Show working and give the answer with unit.
Plot/Sketch	Represent data or key features correctly.
Compare	Give relevant similarities/differences.
Predict/Suggest	Give a scientifically reasonable outcome.
Justify/Evaluate	Support a conclusion using principles or evidence.

Feature Key

- REMEMBER — key fact or relationship.
- HINT — guidance without giving away the full solution.
- COMMON ERROR — a frequent misconception.
- PRACTICAL / INQUIRY — investigation and experimental skills.
- APPLICATION — Physics in everyday life, technology or environment.
- THINK BIG — deeper reasoning.
- EXAM FOCUS — assessment technique.
- EXTENSION / ENRICHMENT — useful material beyond the clearly established core pathway.

Contents

- PART I — Topical Revision & Quizzes: Chapters 1–13
- PART II — Mixed Revision Exercises 1–5
- PART III — Final National-Assessment-Style Practice Tests 1–5
- PART IV — Answers & Marking Guidance

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PART I — TOPICAL REVISION & QUIZZES

Revise one concept at a time. Build confidence, then apply.

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CHAPTER 1 — FLUID FLOW

Quick Revision

- A fluid is a substance that can flow.
- Volume flow rate: $Q = V/t = Av$.
- For steady incompressible flow: $A_1v_1 = A_2v_2$.
- Streamline flow is orderly; turbulent flow is irregular.
- Viscosity is internal resistance to flow.
- For suitable steady-flow conditions, increased speed may correspond to lower static pressure.

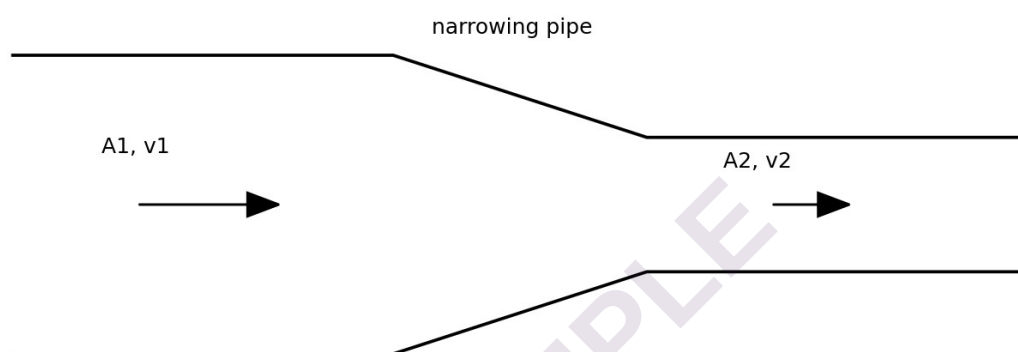


Figure 1.1 Fluid Flow — key scientific representation.

Topical Quiz

30 Marks | Suggested Time: 40 Minutes

1. Define volume flow rate. [2 marks]
2. State the continuity equation for steady incompressible flow. [1 mark]
3. Distinguish streamline from turbulent flow. [2 marks]
4. Define viscosity. [1 mark]
5. Water flows at $6.0 \times 10^{-3} \text{ m}^3 \text{ s}^{-1}$ through area $2.0 \times 10^{-3} \text{ m}^2$. Calculate speed. [3 marks]
6. The pipe narrows to $1.0 \times 10^{-3} \text{ m}^2$. Calculate the new speed. [3 marks]
7. For suitable horizontal steady flow, explain the pressure change in the narrower section. [2 marks]
8. Describe a fair investigation comparing flow through tubes of different cross-sectional area, identifying variables, measurements, repeats and one safety precaution. [8 marks]
9. Explain why mineral deposits narrowing a pipe can increase flow speed for approximately steady incompressible flow. [4 marks]
10. State two practical applications of fluid-flow ideas. [4 marks]

Total: 30 marks

CHAPTER 2 — LINEAR MOTION

Quick Revision

- Distance is scalar; displacement is vector.
- Speed = distance/time; velocity = displacement/time.
- Acceleration $a = (v-u)/t$.
- SUVAT equations apply to constant acceleration.
- Gradient of a velocity-time graph gives acceleration; area gives displacement.
- Negative acceleration does not automatically mean slowing down.

Time / s	0	2	4	6	8
Velocity / m s^{-1}	0	4	8	8	4

COMMON ERROR Negative acceleration does not automatically mean slowing down; compare acceleration direction with velocity.

Topical Quiz

30 Marks | Suggested Time: 40 Minutes

- Distinguish distance from displacement. [2 marks]
- A car changes velocity from 12 to 24 m s^{-1} in 4 s . Calculate acceleration. [2 marks]
- State what the gradient and area of a velocity-time graph represent. [2 marks]
- State two equations of uniformly accelerated motion. [2 marks]
- Explain why negative acceleration does not always mean slowing down. [2 marks]
- A body starts from rest and accelerates at 3 m s^{-2} for 4 s . Find final velocity and displacement. [4 marks]
- Using the data below, plot velocity against time and determine acceleration for the first 4 s , the constant-velocity interval and displacement in the first 6 s . [8 marks]
- A cyclist moving east slows from 10 m s^{-1} to 4 m s^{-1} . Describe the signs of velocity and acceleration if east is positive. [4 marks]
- Give two real-life applications of motion graphs. [4 marks]

Total: 30 marks

Learner workspace / graph or diagram area:

CHAPTER 3 — PROJECTILE MOTION

Quick Revision

- Resolve initial velocity: $u_x = u \cos \theta$, $u_y = u \sin \theta$.
- Horizontal acceleration is zero in the ideal model; vertical acceleration is $-g$.
- At maximum height $v_y = 0$ but acceleration remains $-g$.
- For equal launch/landing levels: $T = 2u \sin \theta / g$, $R = u^2 \sin 2\theta / g$.
- Horizontal and vertical motions are analysed independently.

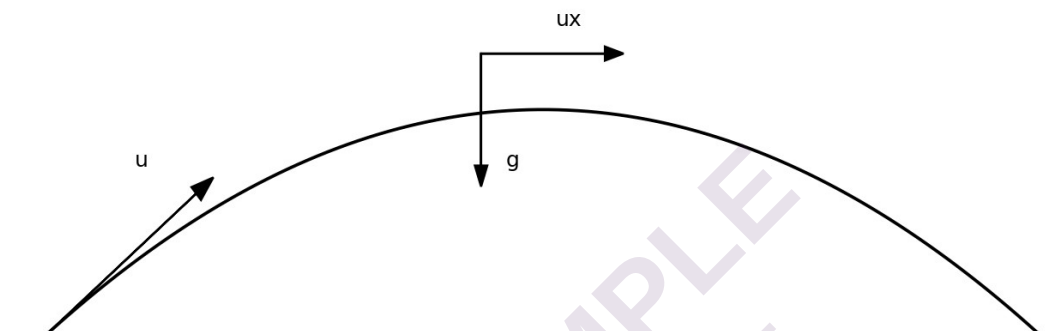


Figure 3.1 Projectile Motion — key scientific representation.

Topical Quiz

30 Marks | Suggested Time: 40 Minutes

20. Resolve a launch speed u at angle θ into horizontal and vertical components. [2 marks]
21. State horizontal and vertical accelerations of an ideal projectile. [2 marks]
22. At the highest point, state v_y and a_y . [2 marks]
23. A ball is projected horizontally at 15 m s^{-1} from 45 m . Take $g=10 \text{ m s}^{-2}$. Find time of flight and range. [4 marks]
24. A football is kicked at 20 m s^{-1} at 30° . Take $g=10 \text{ m s}^{-2}$. Find u_x , u_y , time to highest point, total time, maximum height and range. [8 marks]
25. Compare the ideal ranges for equal launch speed at 30° and 60° . [4 marks]
26. Explain why a ball and a heavier metal object projected horizontally under the same ideal conditions have the same time of fall. [4 marks]
27. Sketch and label an angled projectile trajectory. [4 marks]

Total: 30 marks

CHAPTER 4 — DIMENSIONAL ANALYSIS

Quick Revision

- Fundamental dimensions include M, L and T.
- $[v]=LT^{-1}$, $[a]=LT^{-2}$, $[F]=MLT^{-2}$, $[E]=ML^2T^{-2}$.
- A valid physical equation must be dimensionally homogeneous.
- Dimensional consistency does not prove an equation is physically correct.
- Dimensionless constants cannot usually be determined by dimensional analysis.

Topical Quiz

30 Marks | Suggested Time: 40 Minutes

28. State the fundamental dimensions of mass, length and time. [3 marks]
29. Give dimensions of velocity, acceleration and force. [3 marks]
30. Test $v = u + at^2$ for dimensional homogeneity. [4 marks]
31. Show that mv^2 has dimensions of energy. [3 marks]
32. For $T = k\sqrt{l/g}$, show that $\sqrt{l/g}$ has dimensions of time. [4 marks]
33. Explain why dimensional consistency does not guarantee physical correctness. [3 marks]
34. State two limitations of dimensional analysis. [4 marks]
35. Classify angle/strain/refractive index as dimensionless quantities and explain. [3 marks]
36. Use dimensions to determine the powers in a simple relation of your choice. [3 marks]

Total: 30 marks

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